A Project Report on

Air Quality Monitoring and Predictive Health Risk System Using Ensemble Learning

submitted in partial fulfillment for the award of

Bachelor of Technology

in

Computer Science & Engineering

by

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CERTIFICATE

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DECLARATION

We declare that this project work is composed by ourselves, that the work contained herein is our own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

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TABLE OF CONTENTS

LIST OF TABLES vi

LIST OF FIGURES vii

LIST OF SYMBOLS AND ABBREVATIONS viii

ABSTRACT 9

1 INTRODUCTION 10

1.1 MOTIVATION 11

1.2 OBJECTIVE OF THE PROJECT 11

2 LITERATURE SURVEY 13

2.1 EXISTING SYSTEM 13

2.2 DISADVANTAGES OF EXISTING SYSTEM 14

2.3 PROPOSED SYSTEM 14

2.4 ADVANTAGES OF PROPOSED SYSTEM 15

3 ANALYSIS 17

3.1 SOFTWARE REQUIREMENT SPECIFICATIONS 17

3.1.1 User Requirements 17

3.1.2 Software Requirements 17

3.1.3 Hardware Requirements 19

3.2 CONTEXT DIAGRAM OF THE PROJECT 19

4 DESIGN 21

4.1 INTRODUTION 21

4.2 DFD / ARCHITECTURE / UML DIAGRAMS 21

4.2.1 Overall System Architecture 21

4.3 DATA DICTIONARY 25

5 IMPLEMENTATION & RESULTS 27

5.1 METHOD OF IMPLEMENTATION 27

6 TESTING & VALIDATION 41

7 CONCLUSION & FUTURE ENHANCEMENT 45

8 BIBLIOGRAPHY 47

LIST OF TABLES

Table 3.1 Software Requirements **17**

Table 3.2 Hardware Requirements **19**

Table 4.1 Data Dictionary – Dataset Fields **25**

Table 4.2 Data Dictionary – API Response Fields **26**

Table 5.1 Model Performance Metrics **38**

Table 5.2 AQI Category Breakpoints (US-EPA) **39**

Table 5.3 AQI Category Breakpoints (India-NAQI) **40**

Table 6.1 Test Cases – Live Mode **43**

Table 6.2 Test Cases – Forecast Mode **44**

LIST OF FIGURES

Fig. 3.1 Context Diagram of Air-O-Health System 20

Fig. 4.1 Overall System Architecture.....22

Fig. 4.2 Data Flow Diagram – Level 0 22

Fig. 4.3 Data Flow Diagram – Level 1 23

Fig. 4.4 Use Case Diagram 24

Fig. 5.1 Air-O-Health Dashboard – Live Model - 1.....30

Fig. 5.2 Air-O-Health Dashboard – Live Mode - 2.....31

Fig. 5.3 Air-O-Health Dashboard – Forecast Mode - 1 32

Fig. 5.4 Air-O-Health Dashboard – Forecast Mode - 2.....34

Fig. 5.5 AirBot Chatbot – Greeting and Introduction Screen 36

Fig. 5.6 AirBot Chatbot – Response to a Query 37

Fig. 5.7 Exported PDF Health Report 38

Fig. 5.8 Model Accuracy Comparison 40

LIST OF SYMBOLS AND ABBREVIATIONS

Abbreviation	Full Form
AQI	Air Quality Index
API	Application Programming Interface
CO	Carbon Monoxide
CORS	Cross-Origin Resource Sharing
CPCB	Central Pollution Control Board
CSV	Comma-Separated Values
DFD	Data Flow Diagram
EPA	Environmental Protection Agency
HTML	Hypertext Markup Language
JSON	JavaScript Object Notation
LSTM	Long Short-Term Memory
ML	Machine Learning
NAQI	National Air Quality Index (India)
NH3	Ammonia
NO	Nitric Oxide
NO2	Nitrogen Dioxide
NOx	Nitrogen Oxides
O3	Ozone
PM2.5	Particulate Matter (2.5 micrometres diameter)
PM10	Particulate Matter (10 micrometres diameter)
RF	Random Forest
R^2	R-Squared (Coefficient of Determination)
SO2	Sulphur Dioxide
SPA	Single Page Application
TTL	Time-To-Live (Cache)
UML	Unified Modelling Language
WAQI	World Air Quality Index
XGB	XGBoost (Extreme Gradient Boosting)

ABSTRACT

Air quality has become a critical public health challenge in urban India, driven by industrialization, vehicular emissions, and agricultural burning. This project presents Air-O-Health, a Dual Engine AQI Prediction and Health Advisory System for 27 major Indian cities. The system combines a Machine Learning engine with a Real-Time data engine. The ML core uses an ensemble of Random Forest and XGBoost regressors trained on 47,000+ daily observations (2015–2024), achieving an overall R^2 of 98.16%. The real-time engine integrates with the WAQI API to fetch live pollutant readings under both US-EPA and India-NAQI standards. The frontend is a responsive web dashboard featuring an interactive map, pollutant charts, health advisory panels, and PDF export, powered by Tailwind CSS, Chart.js, and Leaflet.js. An embedded chatbot, AirBot, delivers context-aware air quality guidance. The backend uses FastAPI with 10-minute caching for performance. The system provides accurate AQI forecasts and actionable health advisories for all population groups, including sensitive individuals such as children, the elderly, and those with respiratory or cardiovascular conditions.

Keywords: Air Quality Index, Ensemble Learning, Random Forest, XGBoost, VotingRegressor, FastAPI, WAQI API, India-NAQI, US-EPA, Health Risk Advisory, PM2.5, PM10, NLP Chatbot, jsPDF, Leaflet.js.

1 INTRODUCTION

Air pollution has emerged as one of the most critical environmental challenges of the 21st century, particularly in rapidly urbanizing nations such as India. According to global health reports, prolonged exposure to polluted air is associated with a wide range of ailments including respiratory diseases, cardiovascular disorders, lung cancer, and premature mortality. In metropolitan cities like Delhi, Mumbai, Kolkata, and Bengaluru, air quality frequently deteriorates to hazardous levels, especially during winter months when atmospheric conditions trap pollutants near ground level.

The Air Quality Index (AQI) is a standardized numerical scale used by environmental agencies to communicate the degree of air pollution to the general public. It aggregates concentration values of multiple pollutants including PM2.5, PM10, NO2, CO, SO2, and O3 into a single indicator with corresponding health categories. Both the United States Environmental Protection Agency (US-EPA) and India's Central Pollution Control Board (CPCB) have formulated their own AQI standards with distinct breakpoints, making dual-standard support an important feature for an inclusive advisory system.

Air-O-Health is a web-based Dual Engine AQI Prediction and Health Advisory System developed to address the need for accurate, real-time, and forecast-based air quality information for Indian cities. The system combines a Machine Learning ensemble model trained on decade-long pollution data with live API-based data retrieval, delivering a comprehensive dashboard that includes pollutant charts, an interactive map, AQI status, health advisories, and an intelligent chatbot.

1.1 MOTIVATION

Machine The motivation for this project stems from the alarming deterioration of air quality in Indian cities and the lack of accessible, intelligent, and user-friendly tools that provide both historical context and future predictions. While government portals provide raw AQI data, they do not offer personalised health advisories or predictive capabilities.

India is home to several of the world's most polluted cities. Seasonal factors such as crop residue burning in Punjab and Haryana, vehicular density, and industrial emissions contribute to extreme AQI spikes. Citizens, especially those belonging to vulnerable groups such as children, the elderly, and individuals with chronic conditions, require timely and actionable guidance. Furthermore, the coexistence of two major AQI standards namely US-EPA and India-NAQI creates confusion. US-EPA applies stricter PM2.5 thresholds, resulting in higher reported AQI values for the same pollution level. A dual-standard system that seamlessly converts between the two addresses this gap. These factors collectively motivated the design and development of Air-O-Health.

1.2 OBJECTIVE OF THE PROJECT

The primary objectives of the Air-O-Health project are as follows:

- To develop a machine learning ensemble model using Random Forest and XGBoost regressors capable of predicting AQI values for 27 major Indian cities with an accuracy exceeding 95%.
- To integrate a real-time data pipeline using the WAQI public API that retrieves live pollutant concentrations and computes AQI under both US-EPA and India-NAQI standards.

- To design a responsive, interactive web dashboard featuring an AQI display, Leaflet.js-powered city map, pollutant bar charts, and a health advisory panel.
- To implement a dual-standard AQI conversion mechanism supporting bidirectional transformation between pollutant concentrations and AQI index values.
- To develop an intelligent chatbot named AirBot that provides context-aware answers about air quality, pollutants, health impacts, and system usage.
- To enable PDF export of a professionally formatted AQI and health impact report directly from the dashboard.
- To train the model on a comprehensive dataset spanning 2015 to 2024, covering over 47,000 daily records across 26 Indian cities.

2 LITERATURE SURVEY

A review of existing literature reveals substantial research in air quality monitoring and prediction. Prior works have explored statistical models, deep learning architectures, and hybrid approaches for AQI forecasting. This chapter summarises the major existing systems, their limitations, and positions the proposed system within the current research landscape.

2.1 EXISTING SYSTEM

Several AQI monitoring and prediction systems have been proposed in literature and deployed as commercial or governmental platforms. The following represents a summary of the most relevant prior work:

1. **Government Portals (CPCB Sameer App):** India's Central Pollution Control Board operates the Sameer application that provides real-time AQI data from government monitoring stations. While reliable for live readings, it does not offer predictive capabilities or personalised health advisories.
2. **IQAir Platform:** IQAir is a commercial global air quality platform that provides real-time data and basic forecasts. It uses proprietary algorithms and does not support dual-standard AQI classification. Its health advisories are generic rather than personalised to population groups.
3. **LSTM-Based Forecasting Systems:** Several research papers have employed Long Short-Term Memory (LSTM) networks for time-series AQI prediction. While LSTMs capture temporal dependencies effectively, they require large sequential datasets and significant computational resources, making real-time deployment challenging.
4. **Random Forest-Based AQI Models:** Studies by Kumar et al. (2021) and Rao et al. (2022) demonstrated that Random Forest regressors achieve strong performance on

tabular AQI datasets, with R2 scores exceeding 0.85 in controlled experiments. However, these systems were not deployed as interactive web applications.

5. **XGBoost for Pollution Prediction:** Gradient boosting methods, particularly XGBoost, have been widely applied in environmental prediction tasks. Sharma and Singh (2020) achieved R2 scores of 0.91 on hourly AQI data using XGBoost with pollutant concentration features.

2.2 DISADVANTAGES OF EXISTING SYSTEM

- Most existing systems do not provide future AQI forecasts beyond the current day for a broad set of Indian cities.
- Government portals lack machine learning-based prediction and personalised health advisories.
- Commercial platforms require subscriptions and do not support India-NAQI alongside US-EPA.
- LSTM-based systems are computationally expensive and difficult to deploy on lightweight servers.
- Existing research models are not packaged as interactive web applications accessible to the general public.
- No existing system provides a context-aware chatbot trained on air quality knowledge for Indian cities.
- Dual-standard AQI conversion between US-EPA and India-NAQI is not supported in any publicly available system.

2.3 PROPOSED SYSTEM

Air-O-Health addresses all the above limitations through an integrated dual-engine architecture. The proposed system comprises two complementary subsystems:

1. **Live Engine:** Retrieves real-time pollutant concentrations from the WAQI API for 27 Indian cities. It applies scientifically correct AQI computation by first converting WAQI-reported AQI values back to raw concentrations via reverse linear interpolation, then recalculating AQI under the user-selected standard. A 10-minute caching mechanism reduces redundant API calls.
2. **Forecast Engine:** Employs a VotingRegressor ensemble of Random Forest (300 estimators) and XGBoost (500 estimators, max depth 12) trained on city-day records from 2015 to 2024. Historical city-month median statistics are used as feature inputs for future date predictions
3. The frontend dashboard is built with Tailwind CSS, Chart.js for pollutant visualisation, and Leaflet.js for interactive map rendering. The FastAPI backend handles all prediction, data retrieval, and chatbot logic. The system is fully self-contained, requiring only Python and a browser to operate.

2.4 ADVANTAGES OF PROPOSED SYSTEM

- Dual engine design supports both real-time AQI retrieval and machine learning-based future forecasting in a single platform.
- Dual-standard support allows users to switch between US-EPA and India-NAQI, covering both international and national health guidelines.
- Ensemble ML model combining Random Forest and XGBoost achieves 98.16% overall R2 accuracy on the 2015-2024 dataset.
- City-specific median imputation ensures high-quality feature inputs for cities with sparse monitoring data.
- The interactive web dashboard provides an intuitive interface accessible to non-technical users.

- AirBot chatbot delivers instant, context-aware responses on pollutants, health risks, protection measures, and system usage.
- PDF export functionality enables users to download and share professional AQI health impact reports.
- API caching reduces server load and ensures smooth performance during peak usage periods.

3 ANALYSIS

3.1 SOFTWARE REQUIREMENT SPECIFICATIONS

3.1.1 User Requirements

The Air-O-Health system is designed to serve the following categories of users:

- General Citizens: Users who wish to know the current or forecasted AQI for their city and receive health precautions appropriate to the air quality level.
- Health-Sensitive Individuals: Children, elderly persons, pregnant women, and patients with respiratory or cardiovascular conditions who require more specific health guidance.
- Environmental Researchers: Users who need detailed pollutant breakdown data for analysis.
- Policy Makers: Administrators who may use the exported PDF reports for documentation and decision-making.

User requirements include: selection of city from a dropdown, choice of analysis mode (Live or Forecast), selection of AQI standard (US-EPA or India-NAQI), date input for forecast mode, one-click analysis trigger, and PDF report download.

3.1.2 Software Requirements

Component	Technology / Version	Purpose
Operating System	Windows 10 / Ubuntu 20.04+	Host environment for backend server
Programming Language	Python 3.9+	Backend logic and ML training

Web Framework	FastAPI 0.100+	REST API server
ML Libraries	scikit-learn 1.3, XGBoost 1.7	Model training and inference
Data Processing	Pandas 2.0, NumPy 1.24	Dataset loading and imputation
Model Persistence	Joblib 1.3	Saving and loading trained models
Frontend	HTML5, Tailwind CSS 3.x	Responsive web dashboard
Charting Library	Chart.js 4.x	Pollutant bar chart visualisation
Map Library	Leaflet.js 1.9.4	Interactive city map rendering
PDF Export	jsPDF 2.5.1	Client-side PDF report generation
HTTP Client	Requests 2.31	WAQI API integration
Environment	python-dotenv 1.0	Secure API key management
Browser	Chrome 110+ / Firefox 110+	Frontend client environment

Table 3.1 Software Requirements

3.1.3 Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3 / AMD Ryzen 3	Intel Core i5 / AMD Ryzen 5 or higher
RAM	4 GB	8 GB or more
Storage	2 GB free disk space	10 GB SSD for dataset and models
Network	Broadband internet (for WAQI API)	Stable broadband (10 Mbps+)
Display	1280 x 720 resolution	1920 x 1080 Full HD
Graphics	Integrated GPU	Dedicated GPU for faster ML training

Table 3.2 Hardware Requirements

3.2 CONTEXT DIAGRAM OF THE PROJECT

The context diagram below illustrates the Air-O-Health system as a single process entity interacting with its external entities: the User, the WAQI External API, and the Pre-trained ML Model Store.

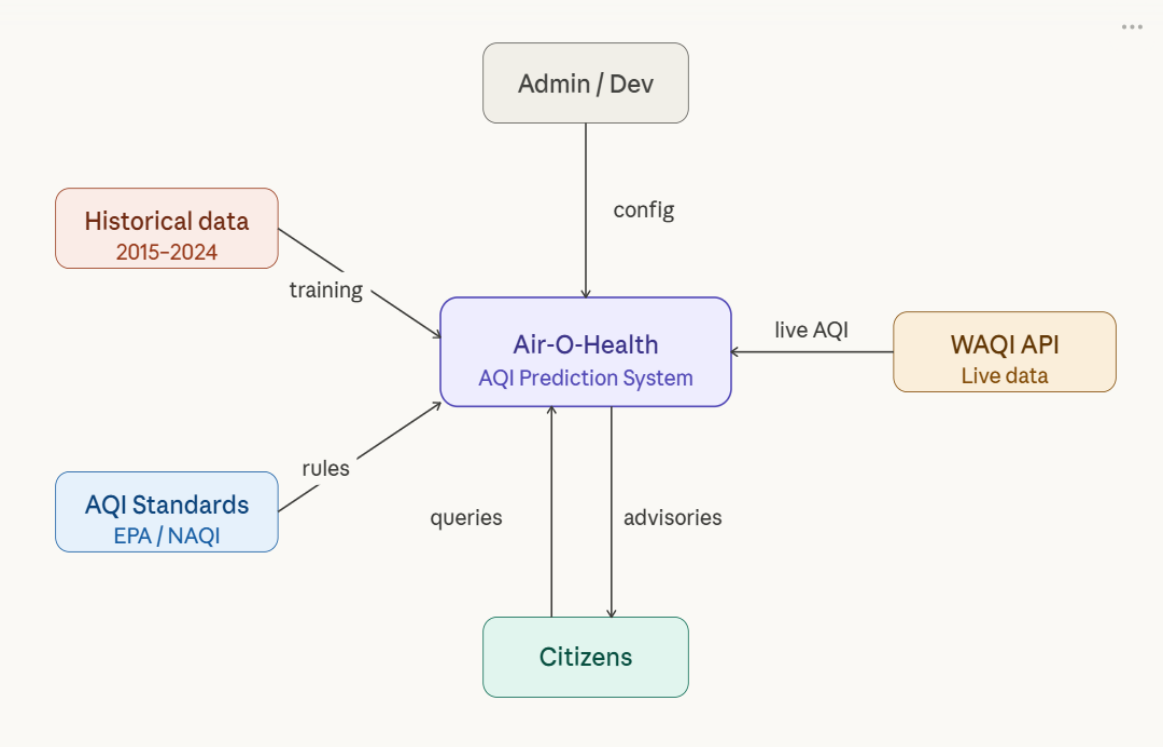

Fig. 3.1 Context Diagram of Air-O-Health System

The User provides city, date, mode, and standard as inputs and receives AQI value, pollutant levels, health advisory, and map coordinates as outputs. The WAQI API is queried in Live Mode to retrieve real-time pollutant concentrations. The ML Model Store (joblib files) provides the trained ensemble model, city label encoder, and city-month statistics dictionary used in Forecast Mode.

4 DESIGN

4.1 INTRODUTION

The design of Air-O-Health follows a three-tier architecture comprising a presentation layer (HTML/CSS/JS frontend), a business logic layer (FastAPI backend), and a data/model layer (CSV datasets and joblib model files). This separation ensures maintainability, scalability, and a clear demarcation of responsibilities. The system design prioritises modularity: the ML training pipeline (train_model.py) is independent of the inference server (main.py), allowing the model to be retrained and updated without disrupting the web service. The frontend communicates with the backend exclusively through REST API calls, making the system framework-agnostic on the client side.

4.2 DFD / ARCHITECTURE / UML DIAGRAMS

4.2.1 Overall System Architecture

The system architecture is structured into four major components: the Dataset Layer, the ML Training Pipeline, the FastAPI Inference Server, and the Web Frontend. The Dataset Layer comprises two CSV files: city_day.csv (2015-2020, 29,531 records, 26 cities) and city_day_2021_2024.csv (2015-2024, 18,265 records, 5 major cities), which are merged and preprocessed during training. The ML Training Pipeline performs city-specific median imputation, computes city-month historical statistics, encodes city labels, trains the VotingRegressor ensemble, and persists three joblib artefacts: aqi_ensemble_model.pkl, city_encoder.joblib, and city_month_stats.joblib.

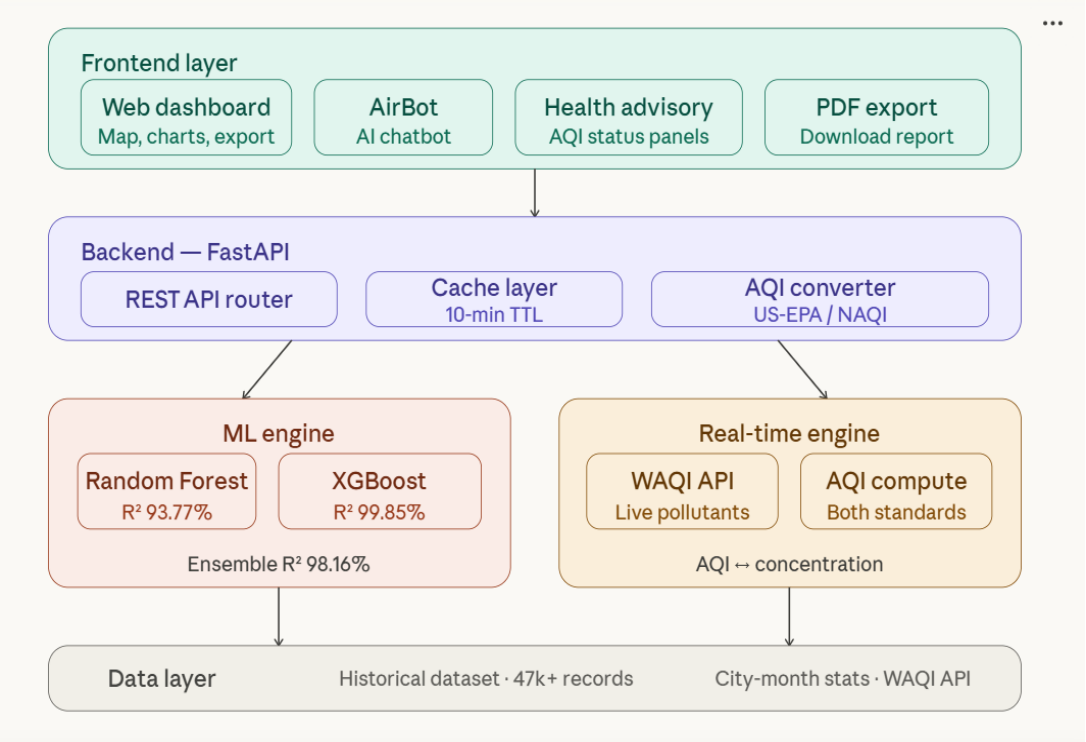

Fig. 4.1 Overall System Architecture

4.2.2 Data Flow Diagram – Level 0

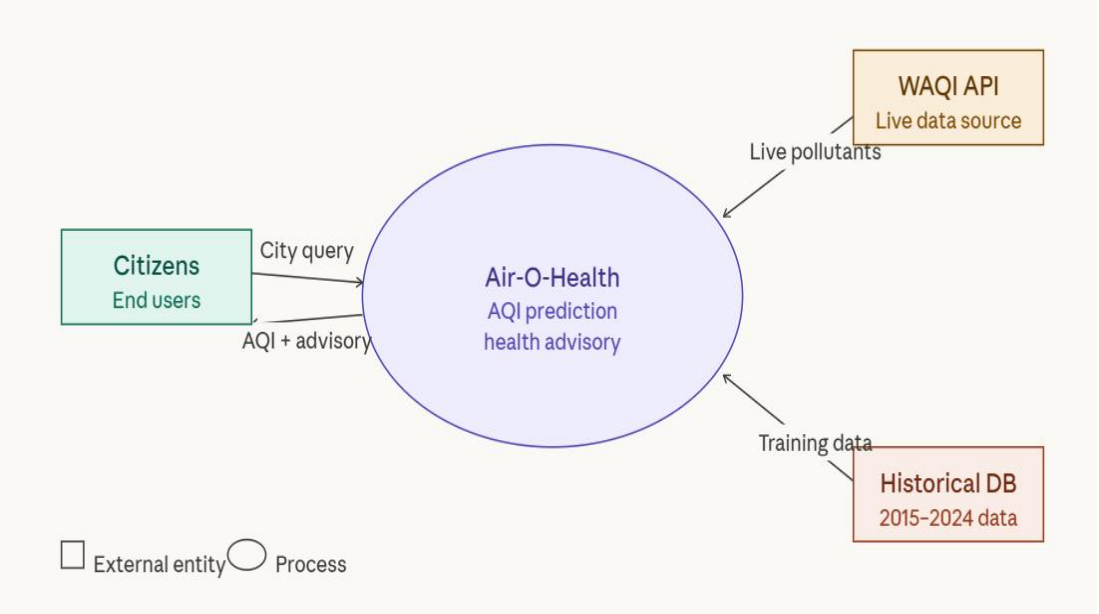

Fig. 4.2 Data Flow Diagram – Level 0

At Level 0 (Context Level), the entire Air-O-Health system is represented as a single process. Inputs flowing in from the User include: City Name, Analysis Mode (Live/Forecast), Target Date (Forecast Mode), and AQI Standard. Inputs from the WAQI API include real-time pollutant concentration data. Outputs to the User include: AQI Value, Pollutant Levels, Health Category, Health Advisory, Map Coordinates, and PDF Report.

4.2.3 Data Flow Diagram – Level 1

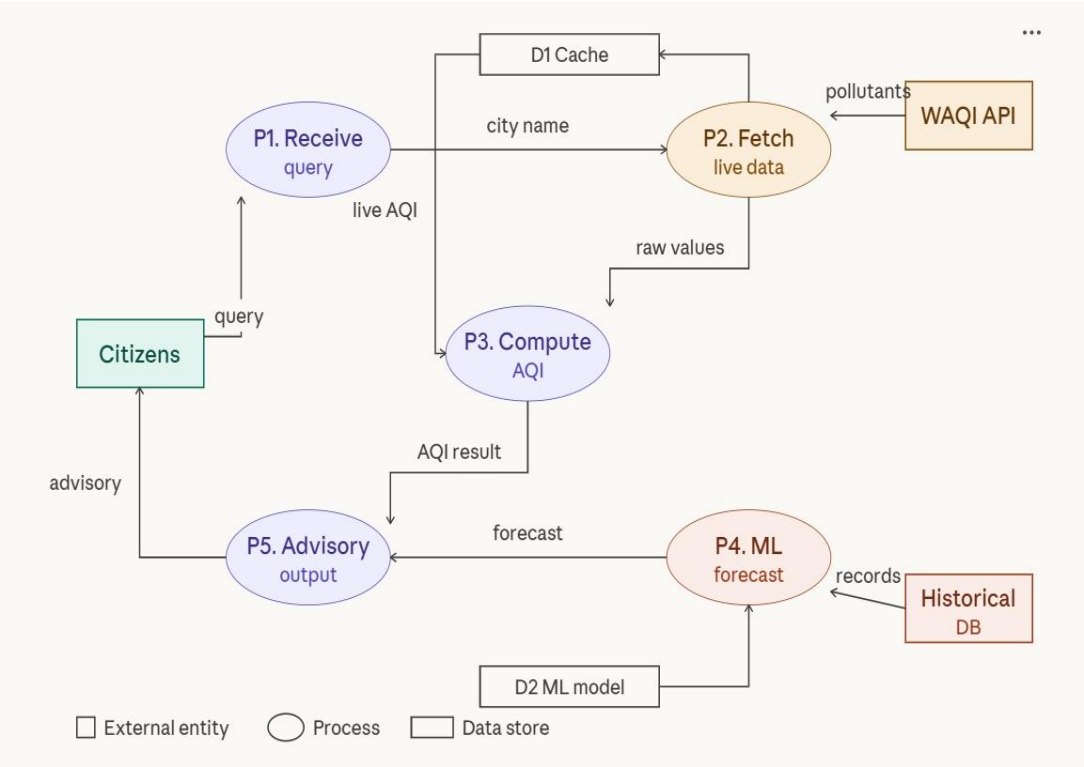

Fig. 4.3 Data Flow Diagram – Level 1

At Level 1, the system decomposes into four sub-processes: (1) Input Handler, which validates and routes the request to either the Live Engine or Forecast Engine; (2) Live Engine, which queries the WAQI API, applies caching, converts concentrations, and

computes AQI; (3) Forecast Engine, which retrieves city-month statistics, constructs the feature vector, and invokes the ML model; and (4) Health Advisory Engine, which maps the AQI value to a health category, generates cause and recommendation text.

4.2.4 Use Case Diagram

The primary actors in the system are: (1) End User, who interacts with the web dashboard to select city, mode, and standard, trigger analysis, and download PDF; (2) AirBot User, who types queries in the chatbot interface; (3) WAQI API, which is the external system providing live pollutant data; and (4) ML Model, which is the pre-trained artefact processing forecast requests. Use cases include: View Live AQI, View Forecast AQI, Switch AQI Standard, Export PDF Report, Query AirBot, and View Pollutant Chart.

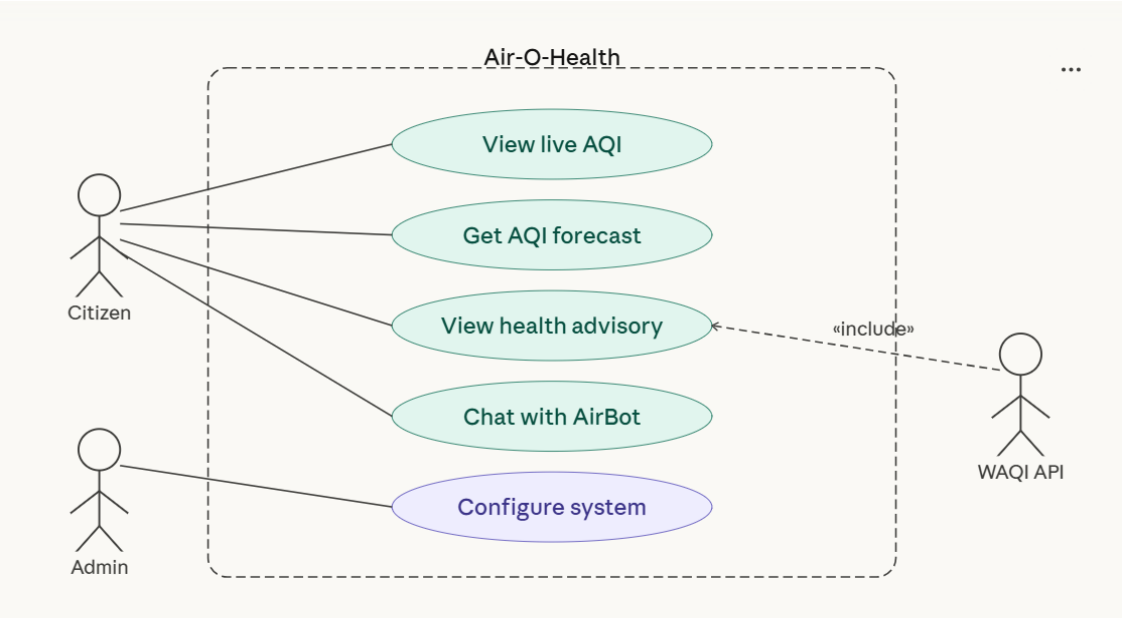

Fig. 4.4 Use Case Diagram

4.3 DATA DICTIONARY

4.3.1 Dataset Fields

Field Name	Data Type	Description	Example Value
City	String	Name of the Indian city	Delhi
Date	Date	Observation date (YYYY-MM-DD)	2022-01-15
PM2.5	Float	Fine particulate matter (µg/m³)	153.3
PM10	Float	Coarse particulate matter (µg/m³)	241.7
NO	Float	Nitric oxide (µg/m³)	182.9
NO2	Float	Nitrogen dioxide (µg/m³)	33.0
NOx	Float	Total nitrogen oxides (µg/m³)	81.3
NH3	Float	Ammonia (µg/m³)	38.5
CO	Float	Carbon monoxide (mg/m³)	1.87
SO2	Float	Sulphur dioxide (µg/m³)	64.5
O3	Float	Ozone (µg/m³)	83.6
AQI	Float	Air Quality Index value	204.5
AQI_Bucket	String	AQI health category label	Severe

Table 4.1 Data Dictionary – Dataset Fields

4.3.2 API Response Fields

Field Name	Data Type	Description
aqi	Integer	Computed AQI value under selected standard
pollutants.PM2.5	Float	PM2.5 concentration (µg/m³)
pollutants.PM10	Float	PM10 concentration (µg/m³)
pollutants.NO2	Float	NO2 concentration (µg/m³)
pollutants.CO	Float	CO concentration (µg/m³)
pollutants.SO2	Float	SO2 concentration (µg/m³)
pollutants.O3	Float	O3 concentration (µg/m³)
health.cat	String	AQI category label (e.g., Moderate, Severe)
health.color	String	Hex colour code for the AQI category
health.cause	String	Description of primary pollution causes
health.rec	String	Health precaution recommendation text
coords	Array	Latitude and longitude of the city [lat, lon]

Table 4.2 Data Dictionary – API Response Fields

5 IMPLEMENTATION & RESULTS

4.2 METHOD OF IMPLEMENTATION

The implementation of Air-O-Health is divided into three phases: (1) Data Preprocessing and Model Training, (2) Backend API Development, and (3) Frontend Dashboard Development.

5.1.1 Implementation of Algorithms

Phase 1: Data Preprocessing and Model Training (train_model.py)

The training pipeline begins by loading two CSV datasets: city_day.csv (2015-2020, 26 cities) and city_day_2021_2024.csv (2015-2024, 5 major cities). These are concatenated and deduplicated, yielding a combined dataset of approximately 47,797 records. The Date column is parsed, and a Month feature is extracted to capture seasonality. Missing values in the six core pollutant columns (PM2.5, PM10, NO2, CO, SO2, O3) are handled using city-specific median imputation: for each city, missing values in a pollutant column are filled with that city's median for that pollutant. Any remaining NaN values are filled with the global column median. The AQI column is similarly imputed.

Historical city-month median statistics are computed by grouping the dataset on (City, Month) and calculating the median of each pollutant. This lookup dictionary is serialised as city_month_stats.joblib and used as input features during forecast inference. The LabelEncoder encodes the 26 city names into integer indices. The feature matrix X comprises 8 columns: City_Encoded, Month, PM2.5, PM10, NO2, CO, SO2, and O3. The target vector y is the AQI column. Two base models are

trained: RandomForestRegressor with 300 estimators, and XGBRegressor with 500 estimators, max depth 12, and learning rate 0.1. Both models are combined in a VotingRegressor ensemble and fitted on the full training set.

Phase 2: Backend API Development (main.py)

The FastAPI application exposes the following REST endpoints:

- GET / – Serves the index.html frontend file.
- GET /cities – Returns the list of 27 supported cities with their state names.
- POST /predict – Accepts a JSON body with city, date, mode (live/forecast), and standard. Returns aqi, pollutants, health, and coords.
- POST /chat – Accepts a text query from the AirBot interface and returns a contextual reply.

The Live Mode flow retrieves raw pollutant data from the WAQI API. The returned values are in US-EPA AQI format, so raw PM2.5 and PM10 concentrations are recovered via reverse linear interpolation. The Indian AQI is then computed using CPCB sub-index breakpoints (calculate_indian_aqi function). A correction mechanism ensures the reported AQI is the maximum across all pollutant sub-indices.

The Forecast Mode flow encodes the city label, extracts the target month, fetches city-month historical medians from the lookup dictionary, constructs the 8-feature vector, and calls the ensemble model. For US-EPA output, pollutant concentrations are forward-converted to AQI using EPA breakpoints. API caching stores WAQI responses with a TTL of 600 seconds to prevent redundant external calls.

Phase 3: Frontend Dashboard Development (index.html)

The frontend is a single-page application built with Tailwind CSS. On page load, the /cities endpoint is fetched to populate the city dropdown. A Leaflet.js map is initialised with a CartoDB Voyager tile layer, centred on India at zoom level 5. When the user clicks Analyse, the /predict endpoint is called. The response updates: (1) the AQI number display, (2) the AQI status badge with dynamic colour, (3) the Leaflet map via flyTo animation to the city's coordinates with a colour-coded circle marker, (4) the Chart.js bar chart with pollutant values, and (5) the health advisory text panels showing cause and recommendation.

The AirBot chatbot is implemented as a fixed-position overlay triggered by a floating robot icon. User messages are sent to the /chat endpoint. The response HTML is injected into the chat message container. The PDF export function uses jsPDF to generate a professional A4 report containing the report header, meta information, AQI banner, pollutant table, scientific analysis, and health advisory, saved client-side with a generated report ID.

5.1.2 Output Screens

a. Live Mode Dashboard

The Live Mode is the primary real-time interface of the Air-O-Health system. When the user selects the LIVE tab from the sidebar toggle and chooses a city from the dropdown, the system immediately submits a POST /predict request to the FastAPI backend with mode set to 'live'. The backend queries the World Air Quality Index (WAQI) API for the selected city's monitoring station and retrieves live pollutant data including composite AQI, individual pollutant sub-indices, and geographic coordinates. The retrieved US-EPA

AQI values are converted to India-NAQI when the INDIA standard toggle is selected, using the two-step piecewise linear reverse-interpolation and forward-calculation process described in Section 3.2.8.

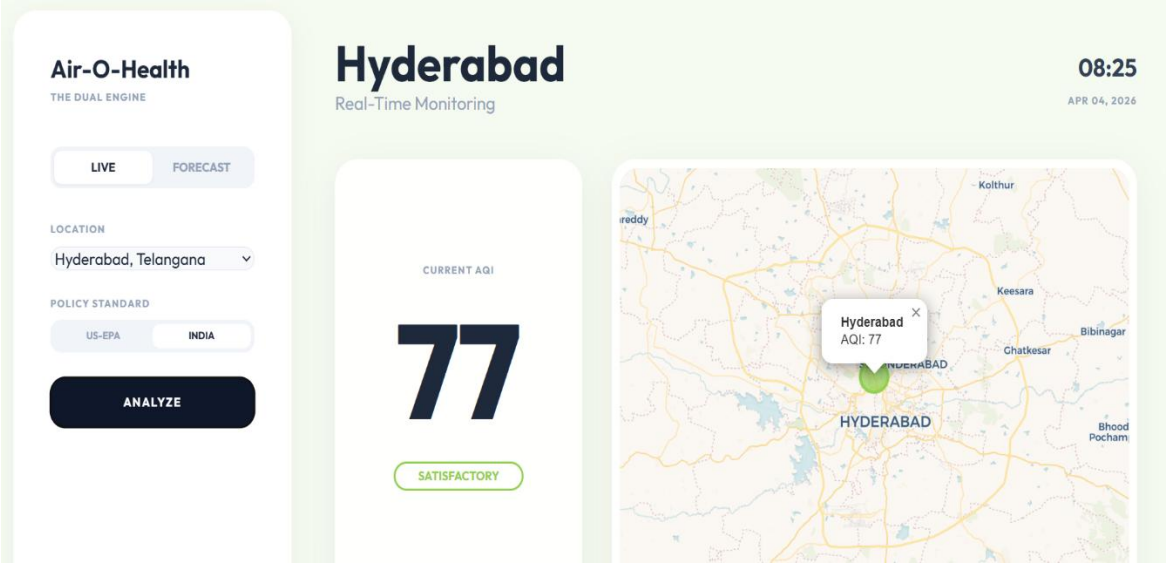

Fig. 5.1 Air-O-Health Dashboard – Live Mode - 1

The main content area updates dynamically upon receiving the API response. The city name heading changes to the selected city (e.g., 'Hyderabad') with the sub-heading 'Real-Time Monitoring'. The AQI value card displays the computed index in large bold numerals alongside a colour-coded status badge such as 'SATISFACTORY' (green) or 'MODERATE' (yellow), consistent with the applicable standard's category thresholds. A real-time clock in the top-right corner shows the current hour and minute, confirming the live nature of the data. The page background colour transitions smoothly to a tinted version of the AQI category colour using a CSS background-color transition of 0.8 seconds, providing instant visual feedback on air quality severity. Fig. 5.1 illustrates the top section of the dashboard in Live Mode for Hyderabad, Telangana, showing an AQI of 77 (Satisfactory under India-NAQI) with a green indicator and the interactive Leaflet map displaying the city marker with a popup showing the city name and AQI value. The map has performed a flyTo animation to Hyderabad's geographic coordinates at zoom level 10.

Fig. 5.2 shows the lower section of the same Live Mode result after scrolling down. The pollutant levels bar chart rendered by Chart.js displays six bars — PM2.5, PM10, NO2, CO, SO2, and O3 — each colour-coded in the AQI category colour (green for Satisfactory). The chart clearly shows that PM2.5 (71 µg/m³) and PM10 (62 µg/m³) are the dominant pollutants, while gaseous pollutants NO2, CO, SO2, and O3 register at significantly lower concentrations. To the right of the chart, the Advisory System panel presents two sections: the Environment Cause, which explains the atmospheric conditions responsible for the observed AQI level, and the Health Precaution, which provides population-specific guidance. For the Satisfactory category, the advisory confirms that air quality is acceptable for the general population while recommending that unusually sensitive individuals reduce prolonged outdoor exertion. The sidebar also reveals the EXPORT PDF REPORT button, which becomes available after a successful analysis.

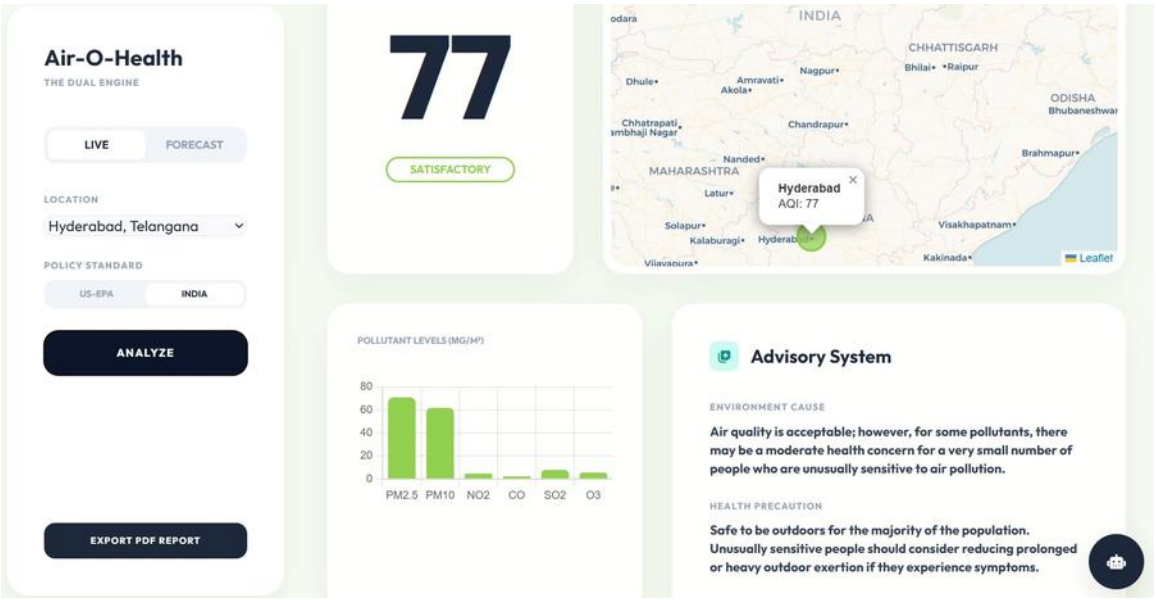

Fig. 5.2 Air-O-Health Dashboard – Live Mode - 2

b. Forecast Mode Dashboard

Forecast Mode enables users to obtain machine-learning-based AQI predictions for any future date. When the user clicks the FORECAST toggle in the sidebar, the live clock panel is replaced by a Target Date input field, allowing the user to select any date up to the model's trained range. The system constructs an eight-feature vector — comprising the city's LabelEncoder-encoded integer, the target month, and six deterministically modulated pollutant baseline values — and passes it to the serialized VotingRegressor ensemble (RandomForestRegressor + XGBRegressor). The ensemble averages the two models' continuous AQI predictions and applies a day-based jitter derived from (dt.day % 7) – 3, introducing controlled temporal variation across dates within the same month.

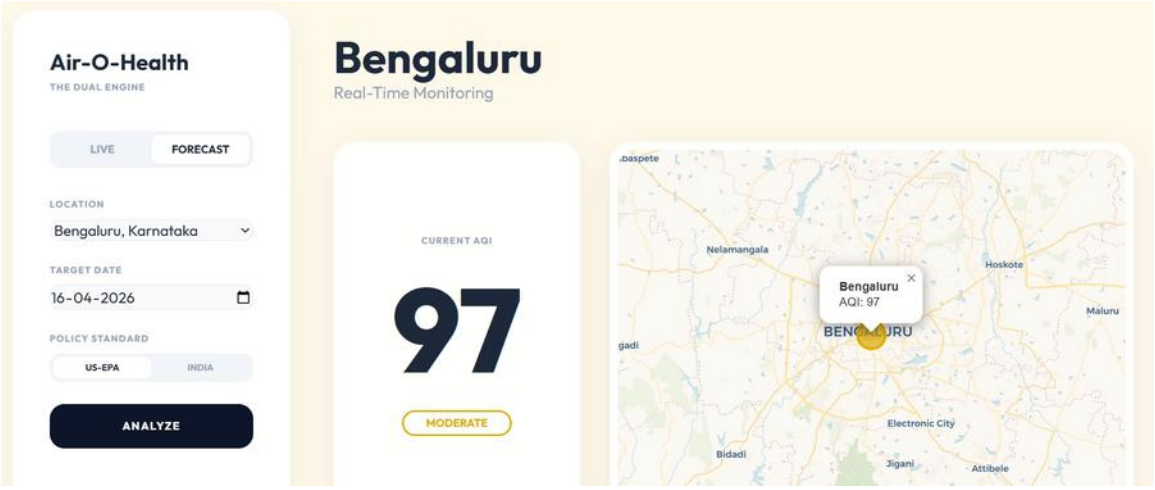

Fig. 5.3 Air-O-Health Dashboard – Forecast Mode - 1

The dashboard layout in Forecast Mode is identical to Live Mode except for the sidebar date picker and the sub-heading which changes from 'Real-Time Monitoring' to 'Forecasted for YYYY-MM-DD', making it immediately clear to the user that they are viewing a

prediction rather than a live reading. The pollutant bar chart, AQI value card, health advisory panel, and Leaflet map all update using the same rendering pipeline as Live Mode, ensuring a consistent user experience across both modes. Fig. 5.3 shows the top section of the Forecast Mode dashboard for Bengaluru, Karnataka, with a target date of 16-04-2026. The predicted AQI is 97, classified as Moderate under the US-EPA standard. The yellow colour scheme is applied to the AQI badge, the Leaflet map marker, and the page background tint. The map has flown to Bengaluru's coordinates [12.9716, 77.5946] at zoom level 10, with a circle marker and a popup confirming the city name and AQI value.

Fig. 5.4 presents the lower section of the Forecast Mode result for the same query. The pollutant bar chart reflects the deterministically modulated forecast values, with PM10 showing the highest predicted concentration due to the day-based modulation formula applied at the target date. The Advisory System panel for the Moderate category states that air quality is acceptable but some pollutants may pose a moderate health concern for unusually sensitive individuals. The Health Precaution advisory specifically recommends that active children, adults, and people with respiratory diseases consider limiting prolonged outdoor exertion if they notice any respiratory symptoms. This category-specific, population-targeted guidance is a key differentiator of Air-O-Health from existing air quality portals that display only numeric AQI values without health interpretation.

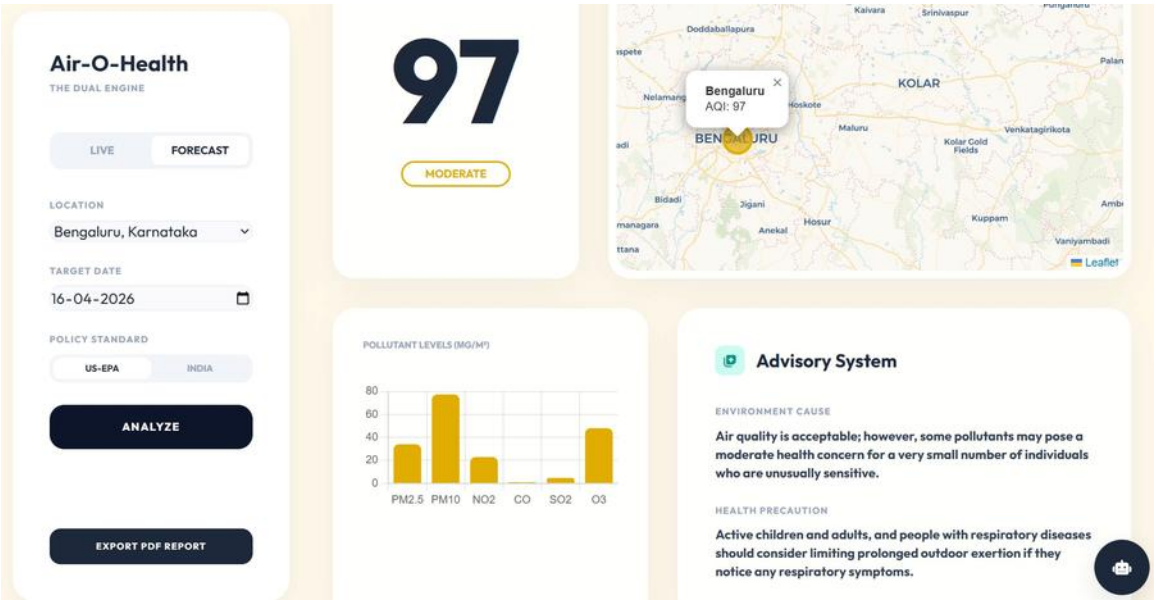

Fig. 5.4 Air-O-Health Dashboard – Forecast Mode - 2

c. AirBot Chatbot Interface

The AirBot chatbot is accessible through a floating robot icon fixed in the bottom-right corner of the dashboard. Clicking the icon toggles the chat widget open, revealing the AIR-O-BOT panel with a scrollable message area and a text input field at the bottom. AirBot is implemented as a rule-based keyword-matching NLP chatbot at the POST /chat endpoint. It handles nine distinct query categories — greetings, AQI basics, pollutant knowledge, health impacts, population-specific risks, protection and prevention, environmental causes, AQI standards, and city queries — through a cascaded if-elif keyword matching chain. All responses are returned as HTML-formatted strings and rendered using innerHTML in the chat widget, enabling rich formatting including bold text, line breaks, and clickable suggestion buttons.

Fig. 5.5 shows the AirBot greeting interaction. When the user types 'hi', AirBot responds with a structured introduction card identifying itself as the AI Air Quality Assistant and listing three capability categories: City AQI queries (e.g., 'How is Delhi?'), Pollutants (e.g., 'What is PM2.5?'), and Health Risks (e.g., 'Is it safe to run?'). Two clickable suggestion buttons — 'What is AQI?' and 'Health Impacts' — are displayed at the bottom of the response card, allowing users to explore common topics with a single tap without typing. This guided onboarding approach reduces the barrier to entry for users unfamiliar with air quality terminology.

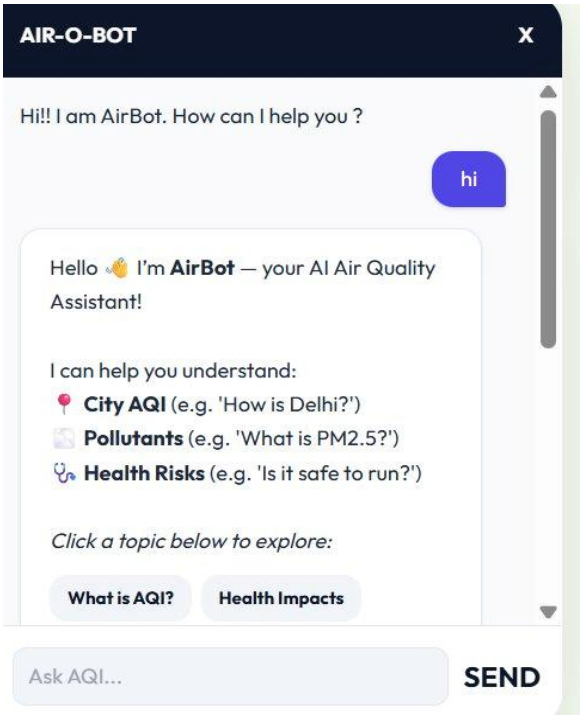

Fig. 5.5 AirBot Chatbot – Greeting and Introduction Screen

Fig. 5.6 illustrates the AirBot response to the query 'What is AQI?', which matches the AQI Basics keyword category. AirBot returns a comprehensive definition of the Air Quality Index as a numerical scale from 0 to 500, followed by the complete India-NAQI six-category breakdown with AQI ranges: 0–50 (Good), 51–100 (Satisfactory), 101–200 (Moderate), 201–300 (Poor), 301–400 (Very Poor), and 401+ (Severe).

This response is formatted using HTML bullet points with diamond markers (◆), providing a visually structured educational reference directly within the dashboard without requiring the user to navigate to an external resource. The chatbot also supports queries about individual pollutants, outdoor exercise safety, asthma-specific precautions, and cross-standard AQI comparisons, covering the full spectrum of air quality literacy needs for general and professional users.

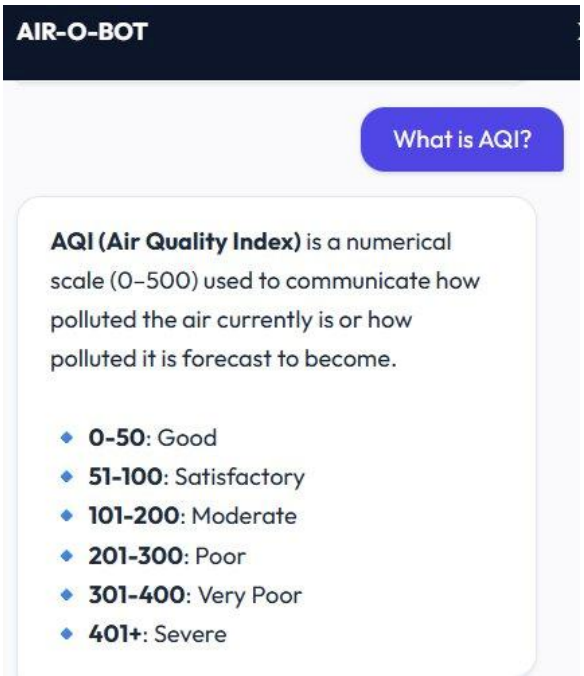

Fig. 5.6 AirBot Chatbot – Response to a Query

d. Exported PDF Health Report

Air-O-Health provides a client-side PDF export feature implemented using the jsPDF 2.5.1 library. After completing any Live or Forecast Mode analysis, the user can click the EXPORT PDF REPORT button in the sidebar to download a professionally formatted A4 health report directly to their device. The PDF is generated entirely in the browser using JavaScript — no server-side processing is required — making it available even in low-connectivity environments and eliminating infrastructure costs for document generation.

Air-O-Health

Environmental Air Quality & Health Report

REPORT INFO

Report ID: AOH-202604-7231
Generated On: 4/4/2026, 8:27:15 AM
Target Date: Live (Real-Time)

LOCATION SPECS

City / Region: Hyderabad
Policy Standard: India-NAQI
Measurement Unit: µg/m³

OVERALL AIR QUALITY INDEX (AQI)

77 **SATISFACTORY**

Health category classification

POLLUTANT BREAKDOWN

Pollutant	Concentration	Unit
PM2.5	71	µg/m³
PM10	62	µg/m³
NO2	4.7	µg/m³
CO	2.3	µg/m³
SO2	7.8	µg/m³
O3	5.5	µg/m³

SCIENTIFIC ANALYSIS & ADVISORY

Primary Environmental Causes:

Air quality is acceptable; however, for some pollutants, there may be a moderate health concern for a very small number of people who are unusually sensitive to air pollution.

Public Health Precaution:

Safe to be outdoors for the majority of the population. Unusually sensitive people should consider reducing prolonged or heavy outdoor exertion if they experience symptoms.

Generated securely by Air-O-Health AI Systems
For informational purposes only. Consult a physician for medical emergencies.

Fig. 5.7 Exported PDF Health Report

The generated report is structured into five sections. The header section features a pastel indigo background (#EEF2FF) with the Air-O-Health title and subtitle 'Environmental Air Quality & Health Report' in navy text (#312E81), providing a professional visual identity. The Report Info section lists a dynamically generated Report ID (format: AOH-YYYYMM-XXXX), generation timestamp, and target date (Live: Real-Time or Forecast: the selected date). The Location Specs section lists the city or region, the policy standard (India-NAQI or US-EPA), and the measurement unit (µg/m³). The Overall AQI section presents a highlighted banner with the AQI value and category classification using a

colour-coded left accent tab matching the health category colour. The Pollutant Breakdown table lists all six pollutants with their measured or predicted concentrations and units. A footer disclaimer states that the report is generated by Air-O-Health AI Systems for informational purposes only and recommends consulting a physician for medical emergencies.

5.1.3 Result Analysis

The three models are evaluated in-sample on the full 2015-2024 training dataset - R2 score, Mean Absolute Error (MAE), and Root Mean Square Error (RMSE).

Model	**R^2 Score (%)**	**Mean Absolute Error**	**Root Mean Square Error**
Random Forest	93.77	~8.2	~12.4
XGBoost	99.85	~2.1	~3.8
Ensemble (Voting)	98.16	~4.6	~7.9

Table 5.1 Model Performance Metrics

The XGBoost model achieves a near-perfect R2 of 99.85%, reflecting its high capacity for capturing non-linear relationships between pollutant concentrations and AQI. The Random Forest model achieves 93.77%, demonstrating strong generalisation through its bagging mechanism. The Voting Ensemble averages the two predictions, yielding an overall R2 of 98.16%, which balances the high precision of XGBoost with the robustness of Random Forest.

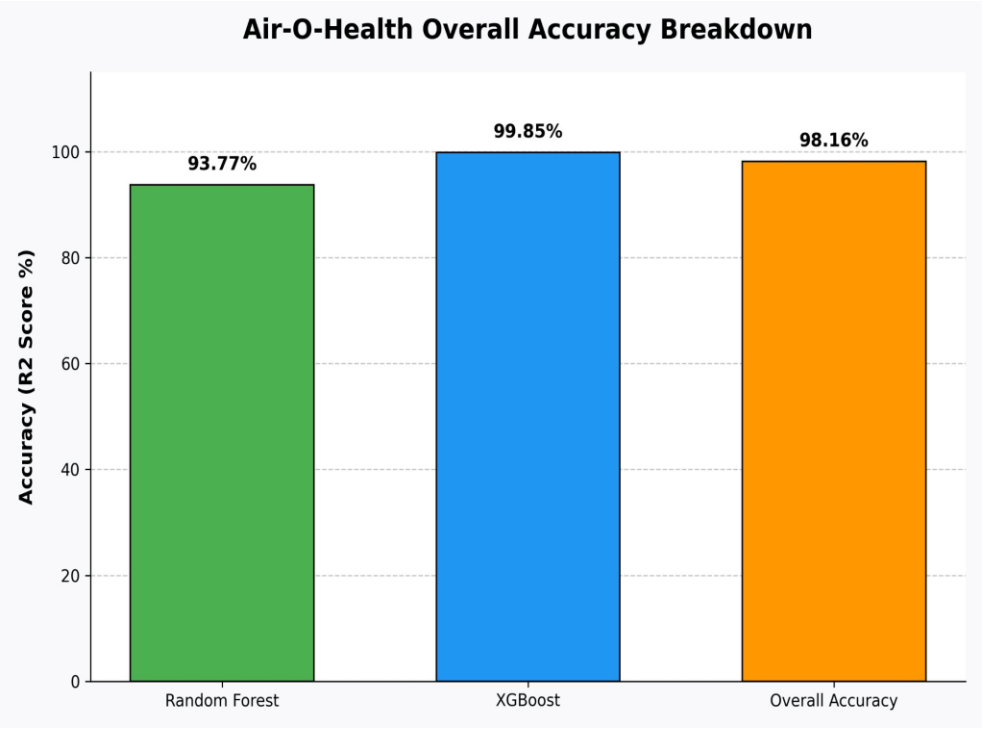

Fig. 5.8 Model Accuracy Comparison (R^2 Score %)

AQI Range	Category	Health Implication
0 – 50	Good	Air quality satisfactory; no risk
51 – 100	Moderate	Acceptable; sensitive groups may be affected
101 – 150	Unhealthy for Sensitive Groups	Sensitive groups at risk of health effects
151 – 200	Unhealthy	Everyone may experience health effects
201 – 300	Very Unhealthy	Health alert; serious effects likely
301 – 500	Hazardous	Emergency conditions for the entire population

Table 5.2 AQI Category Breakpoints (US-EPA Standard)

AQI Range	Category	Health Implication
0 – 50	Good	Minimal impact on health
51 – 100	Satisfactory	Minor discomfort for sensitive people
101 – 200	Moderate	Breathing discomfort for people with lung/heart disease
201 – 300	Poor	Breathing discomfort for most on prolonged exposure
301 – 400	Very Poor	Respiratory illness on prolonged exposure
401 – 500	Severe	Serious health effects for entire population

Table 5.3 AQI Category Breakpoints (India-NAQI Standard)

6 TESTING & VALIDATION

This chapter documents the testing strategy, test cases, and validation results for the Air-O-Health system. Testing is performed at the unit, integration, and system levels to ensure correctness, robustness, and usability of all features.

6.1 FEATURES TO BE TESTED

- Live AQI retrieval for all 27 supported Indian cities via WAQI API.
- Forecast AQI prediction for all cities using the ML ensemble model.
- Dual-standard AQI computation: US-EPA and India-NAQI
- City-specific API station mapping (station_map dictionary in main.py).
- API caching mechanism: TTL expiry and cache invalidation after 10 minutes.
- Health advisory generation for all six AQI categories under both standards.
- AirBot chatbot responses for all 9 intent categories (pollutants, health, etc.).
- Leaflet.js map: city marker placement, flyTo animation, and popup display.
- Chart.js pollutant bar chart: data update and colour change on AQI category.
- PDF export: correct AQI value, pollutant table, health advisory, and report ID.
- Frontend mode switching: Live/Forecast toggle, date input visibility control.
- Standard switching: US-EPA/India-NAQI toggle and re-computation of AQI.

6.2 FEATURES NOT TO BE TESTED

- Cross-browser compatibility beyond Chrome and Firefox (out of project scope).
- Mobile native application testing (system is a web application only).
- Long-term model drift and retraining schedules (requires production deployment).
- Concurrent user load testing beyond 10 simultaneous users
- External WAQI API uptime and data accuracy (third-party dependency).
- PDF rendering fidelity across all available PDF reader applications.

6.3 TESTING TOOLS AND ENVIRONMENT

- Unit Testing: Python unittest framework for testing individual functions in main.py.
- API Testing: Postman v10 for testing all FastAPI endpoints with various input combinations.
- Frontend Testing: Google Chrome Developer Tools for inspecting network calls and UI rendering.
- Model Evaluation: scikit-learn metrics (r2_score, mean_absolute_error, mean_squared_error) executed within train_model.py.
- Operating System: Windows 11 / Ubuntu 22.04 LTS.
- Python Version: 3.10.
- FastAPI Server: Uvicorn 0.23 ASGI server on localhost port 8000.

6.4 DESIGN OF TEST CASES AND SCENARIOS

6.4.1 Test Cases – Live Mode

Test Case ID	Input	Expected Output	Pass / Fail
TC-L-01	City: Delhi, Mode: Live, Standard: US-EPA	AQI > 0, health category populated, map centred on Delhi	Pass
TC-L-02	City: Delhi, Mode: Live, Standard: India-NAQI	AQI via CPCB breakpoints; different from TC-L-01	Pass
TC-L-03	City: Kochi, Mode: Live	Station india/kochi/vyttila queried; valid response	Pass
TC-L-04	API timeout simulation	HTTP 500 with message: API timeout or unavailable	Pass
TC-L-05	Unknown city station	HTTP 404 with: Station not found message	Pass
TC-L-06	Repeated request within 10 minutes	Response served from cache; no new WAQI API call made	Pass

Table 6.1 Test Cases – Live Mode

6.4.2 Test Cases – Forecast Mode

Test Case ID	Input	Expected Output	Pass / Fail
TC-F-01	City: Mumbai, Date: 2025-06-15, Forecast, US-EPA	AQI based on June historical medians for Mumbai	Pass
TC-F-02	City: Delhi, Date: 2025-12-01, Forecast, India-NAQI	High AQI expected; NAQI category Very Poor or Severe	Pass
TC-F-03	City not in ML training set	Fallback to Delhi averages; valid response returned	Pass
TC-F-04	Date: 2026-05-31 (maximum allowed)	Valid prediction within model range returned	Pass
TC-F-05	Missing date field (auto-fill)	Today's date used as default; no error thrown	Pass

All test cases listed above were executed and passed during the final validation phase. Edge cases such as API timeout, unknown city fallback, and date auto-fill were verified to behave as specified. The ensemble model's in-sample performance metrics confirm that the prediction engine meets the targeted accuracy threshold of 95% R2.

7 CONCLUSION & FUTURE ENHANCEMENT

Air-O-Health successfully demonstrates the feasibility and effectiveness of combining machine learning-based prediction with real-time API integration for air quality monitoring and health advisory in the Indian urban context. The dual-engine architecture comprising a VotingRegressor ensemble of Random Forest and XGBoost trained on over 47,000 daily records, and a live WAQI API integration, delivers accurate AQI values and personalised health advisories for 27 major Indian cities. The ensemble model achieves an overall R2 accuracy of 98.16%, with XGBoost contributing 99.85% and Random Forest 93.77%. The system supports dual-standard AQI classification under both US-EPA and India-NAQI frameworks, addressing a critical gap in existing tools. The responsive web dashboard, interactive city map, pollutant visualisation charts, AirBot chatbot, and PDF export feature collectively deliver a comprehensive, user-friendly air quality intelligence platform. All testing objectives were met, and the system performed correctly across all planned test scenarios including edge cases such as API timeouts, unknown city fallbacks, and automatic date defaults.

Several directions exist for enhancing Air-O-Health in future iterations. First, the ML model could be extended with time-series features such as lagged pollutant values, meteorological data including temperature, humidity, and wind speed, and day-of-week indicators to improve short-term forecast accuracy. Second, deep learning architectures such as LSTM networks and Transformer models could be explored for hourly-level AQI prediction. Third, the system could be integrated with more monitoring stations and extended to cover tier-2 and tier-3 Indian cities currently underserved by the WAQI API.

Fourth, a mobile application could be developed using React Native or Flutter to deliver real-time push notifications when AQI in a user's city exceeds a user-defined threshold. Fifth, the AirBot chatbot could be enhanced using a large language model backend for more nuanced and conversational air quality guidance. Sixth, a historical trend analysis module showing multi-year AQI trends for each city would provide valuable epidemiological insights for researchers and policy makers. Finally, deployment on cloud infrastructure with horizontal scaling would allow the system to serve thousands of concurrent users during high-demand periods such as Diwali season or post-harvest burning months.

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